

MX17 drift window & drift voltage

measurement-driven sizing of the DREAM readout for beam data

Dylan Neff

2026-07-19 (run_54 cosmunics + run_55 beam, nTOF EAR2)

TL;DR: latency 35 → **32**, n_samples 32 → **28**, det A drift 600 → **700 V** (staged), B/C/D stay 800 V, short on-beam det-A drift scan as the 700 V spark gate.

The question and the constraints

Decide for the 4-det Micromegas stack:

- ▶ minimum `n_samples` (60 ns DREAM sampling) with no drifting primary lost on *any* detector;
- ▶ drift HV: det A spark-limited (sparked @800 V, stable @600 V) — push to 700?

Hard facts

- ▶ 30 mm gap, Ar/iC₄H₁₀ 90/10, all four dets;
- ▶ **series gas line A→B→C→D** (wetness rises down-chain);
- ▶ current: $32 \text{ smp} \times 60 \text{ ns} = 1.92 \text{ } \mu\text{s}$, latency 35;
- ▶ drift: A 600 V, B/C/D 800 V; resist 560 V.

Data used (2026-07-18)

- ▶ run_54: cosmics, 4 subruns, 9k evts/FEU;
- ▶ **run_55: beam** (19:11, Mode-3 scint-doubles, r560 subruns) — real trigger, real flash;
- ▶ Magboltz: clean 90/10 @CERN P + H₂O grid.

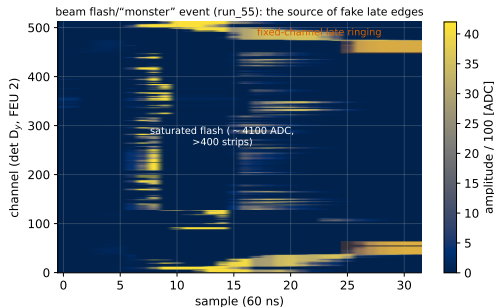
June lesson: wet gas ($\sim 1\%$ H₂O) *halves* low-field drift speed; the 500 V June point truncated its window.

Method: de-noising is the whole game

Extractor (24_drift_time_edges.py)

- ▶ common mode per 64-ch DREAM chip (noise is per-connector), 2 iterations, track strips masked;
- ▶ per-strip robust MAD threshold + width ≥ 2 , hot-channel mask;
- ▶ spatial clusters (≥ 3 adjacent strips) = track;
- ▶ per event: prompt (earliest peak), deep edge (latest peak), pulse rise/end, span.

The decisive cut: events with >100 hit strips (“monster” /flash, $\sim 2\%$ of beam triggers; det-D sparks in cosmoics) are excluded.



Finding 1 — the “slow/late B & D” of the first look was artifact

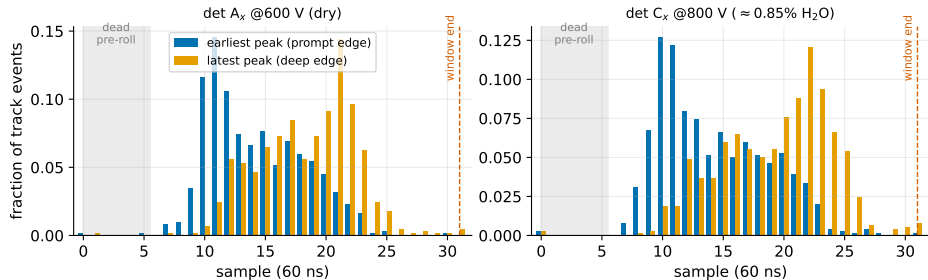
- ▶ Saturated flash events are followed by **fixed channel blocks** (e.g. FEU2 ch 38–63, 448–472) ringing at samples 24–31 $\Rightarrow$ they masqueraded as 25-strip “late clusters”. Every inspected late cluster was one of these.
- ▶ det D cosmics: 129–152 spark events / 9000 $\Rightarrow$ after the cut, D_x ceiling fraction 8.9% $\rightarrow$ 1.5%.
- ▶ det B looks contaminated only in the cosmic run (coherent noise); its *beam* x-plane matches C/D. No evidence B is slow.

ceiling fraction (peak $\geq$ smp 31)	A_x	C_x	D_x	B_x
raw first look (handoff)	0.3%	0.6%	7–20%	21–30%
de-noised + monster cut (cosmics)	0.3%	0.4%	1.5%	(noise-limited)

No detector is losing drifting primaries at 32 smp / latency 35 today.

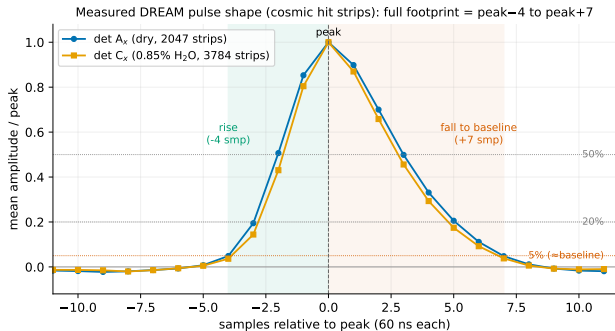
Finding 2 — timing is universal: prompt at sample 8–9, both triggers

run_54 cosmoics, de-noised track clusters — latency 35



- First signal above threshold: sample 6–7 (p5); prompt *peak*: 8–9 $\Rightarrow$ peak $\approx$ latency–26.
- Identical in cosmic-singles and beam Mode-3 triggers, all four detectors — the suspected per-detector t_0 offsets were cluster-geometry artifacts. **One latency serves both run types.**
- **These are PEAK positions.** The pulse has width — it rises 4 smp before the peak and falls +7 smp after (next slide). The window must hold the rise (with baseline) AND the fall, not just the peaks.

Pulse footprint — the window must hold rise + peak + fall

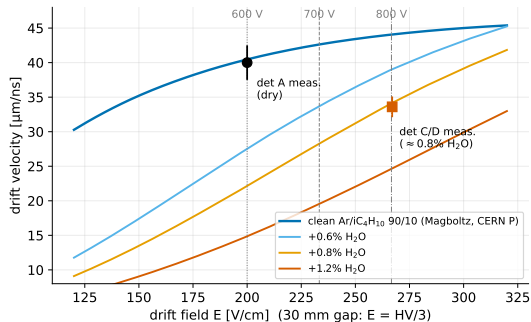


Peak-aligned mean pulse (cosmic hit strips, A_x & C_x identical):

- rise onset (5%) at **peak-4**; want ~ 3 baseline before it.
- fall to 20% at **peak+5**; to baseline (5%) at **peak+7**.

A primary spans ~ 11 smp (peak-4 $\rightarrow$ peak+7). The earlier “+2 fall” was wrong by ~ 5 smp — that is what moves the trim from $n=28$ back to $n=30$.

Finding 3 — water: A is dry; B, C, D all breathe $\sim 0.8\%$ H_2O

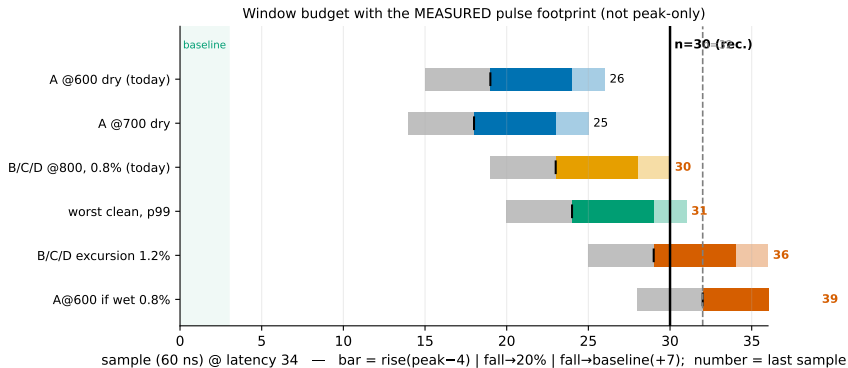


Span of ≥ 6 -strip clusters (cosmics, p95), recorded-column fraction 0.89 anchored on det A = clean:

det	HV [V]	T_{full} [smp]	v [$\mu\text{m/ns}$]	H_2O
A	600	12.4	≈ 40	$\leq 0.2\%$
C	800	15.2	33.0	0.85%
D	800	14.6	34.2	0.80%
B	800	~ 15	~ 33	$\sim 0.8\%$

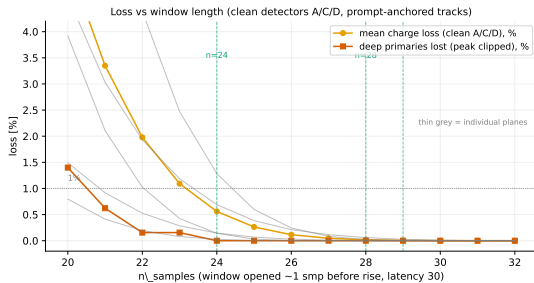
- Chain wetting **saturates at the first hop**: no $A \rightarrow D$ slope, $B \approx C \approx D$ near June's $\sim 1\%$ equilibrium.
- Deep-charge attenuation mild: $\lambda_{\text{eff}} \approx 23\text{--}25 \text{ mm}$ (June: 16–18).

Window budget (measured pulse footprint)



- ▶ A primary spans rise(peak-4) → fall(peak+7). Full containment of every clean-det primary ⇒ latency 34 / n=30 (only 2 smp trim).
- ▶ But we don't need every tail: the next slide prices the *loss* of cutting harder.

How aggressively can we cut? — quantified loss (clean dets A/C/D)



Loss = measured pulse shape (charge vs truncation point) folded on the deep-edge distribution, prompt-anchored:

- ▶ 0 % of primaries lost down to $n=24$ — peaks are always captured; only the far charge *tail* is clipped.
- ▶ mean charge loss < 0.6 % to $n=24$; steep knee below $n \approx 23$.
- ▶ opening 1 smp before the rise costs a fixed ~ 1 % (anomalously early risers, likely not clean tracks).

option	lat	n	cut
conservative	34	32	0
balanced (now)	32	29	3
aggressive	31	28	4
max-trim (B cleared)	30	24	8

Gate = det B (piles at smp 31): all but max-trim keep that bin. **Now: latency 32, $n=29$ (cut 3, ~ 0 loss). After clearing B: latency 30, $n=24$ (cut 25%, $< 0.6\%$ charge).**

Det A: 600 $\rightarrow$ 700 V (staged) — insurance, not speed

Why

- ▶ Today A is dry: 700 V buys only 0.7 smp. Irrelevant.
- ▶ **The one catastrophic corner:** A@600 V in gas as wet as B/C/D *already are* (0.8%) $\Rightarrow$ 22.6 smp column $\Rightarrow$ truncates any affordable window (June's 500 V failure mode).
- ▶ A@700 V in the same gas: 17.7 smp $\Rightarrow$ fits $n=28$.
- ▶ A is first in the chain, but one flow drop / line disturbance makes it wet overnight.
- ▶ Physics cost $\approx$ nil (micro-TPC z-sampling -6%).

How (spark risk: sparked @800 V)

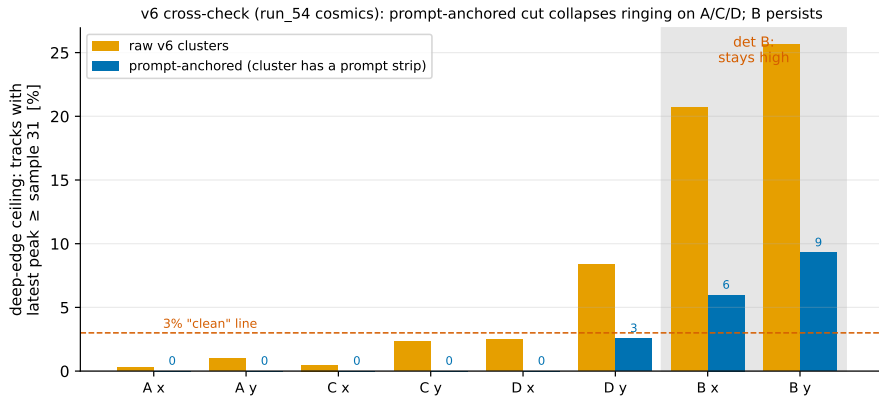
1. one subrun at **650 V**, watch `hv_monitor.csv` (trips, $I > 0.2 \mu\text{A}$);
2. then **700 V** soak under beam;
3. fallback 650 V (still halves the wet-corner risk).

B/C/D: keep 800 V — wetter gas needs the field; nothing gained by lowering.

Proposed on-beam drift scan (small, cheap, do it)

- ▶ **det A only:** 600 V (reference exists) $\rightarrow$ 650 $\rightarrow$ 700, ≥ 30 min each during normal Mode-3 running. B/C/D untouched $\Rightarrow$ their physics data continues.
- ▶ Yields per point: ~ 100 – 200 clean A-tracks/plane $\rightarrow$ span to ± 1 smp.
- ▶ Deliverables:
 1. **spark/current stability at 650/700 = the actual gate** for the HV move;
 2. span(600)/span(700) ratio pins A's water content assumption-free (dry: 1.06, 0.8%-wet: 1.28 — cleanly separable);
 3. confirms the deep-edge + fall endpoint before freezing latency 34 / n=30.
- ▶ Optional, low priority: one B/C/D point at 750 V (wet-gas $v(E)$ slope); costs their margin while running.

v6 cross-check (done locally) — A/C/D confirmed, det B flagged



Prompt-anchored cut (track cluster must contain a near-mesh strip, earliest $\leq$ smp 12) removes ringing clusters that lack one:

- **A, C, D collapse to 0–2.6%** — deep edges at smp 19–24, safely inside $n=28$. D_y's 8.4% was ringing, as predicted.

- **det B stays 6–9%** at the smp-31 ceiling even anchored (span $\approx 20 \rightarrow v \approx 25 \mu\text{m/ns}$). Small counts, cosmic-only.
- Either real slow drift (B wetter) or residual coherent noise — unresolved.
- **Impact:** B's fall rides the ceiling: *clear B*

Summary & open items

knob	recommendation
latency	35 → 32 now → 30 after clearing B
n_samples	32 → 29 now (cut 3, ~0 loss) → 24 after B (cut 25%, <0.6% charge)
det A drift	700 V, staged 650 → 700 with HV-monitor soak
B/C/D drift	keep 800 V
beam drift scan	yes — det A 600/650/700, 30 min each

- ▶ Loss priced from the **measured pulse** (rise -4, fall +7): 0% primaries lost, <0.6% charge to n=24 on clean det. **det B** (piles at smp 31) is the only gate to max-trim.
- ▶ v6 cross-check **DONE** locally (ROOT off-box; **never heavy jobs on the DAQ**): A/C/D clean, **det B flagged**.
- ▶ Electronics follow-up: post-saturation ringing blocks (fixed per-FEU channel ranges); window trim already cuts most of it offline.
- ▶ Full writeup: `mx_july_beam_qa/DRIFT_WINDOW_ANALYSIS.md` (§6 = DAQ OOM incident).